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Quantitative Methods for Humanities

The Linear Regression Model

2024/2025

R Requirements

Take lessons PREDICTION1, PREDICTION2, and PREDICTION3 from course `qss-swirl`

The Linear Model

What is Linear Regression?

- A statistical method to model the relationship between a **dependent variable** (Y) and one or more **independent variables** ($X_1, X_2, \ldots, X_p$).
- **Simple linear regression:** models the relationship between Y and a single predictor (X).
- **Multiple linear regression:** involves two or more predictors.
- Assumes a **linear relationship between the predictors and the average of the dependent variable.**

Model Equation

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

Why Linear Models?

- Linear models provide a simple yet powerful framework for **prediction.**
- When managed properly, it is also useful for **causal inference.**
- Easy to understand and interpret.
- Fast to compute.
- Serves as a foundation for more complex models.

The Multiple Linear Regression

Given a sample of the response Y and the regressors $X_1, \ldots, X_k$, **the MLR model takes the form**

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik} + u_i, \quad i = 1, \ldots n,$$

where

- Y_i is the ith observation of the random (**response**) variable Y.
- X_{ij}, is the ith observation of the random (**regressor**) variable X_j, for $j = 1, \ldots, k$.
- $\beta_0, \beta_1, \ldots, \beta_k$ are the **coefficients** of the regression.
- β_j, for $j = 1, \ldots, k$, are the **slope** coefficients of $X_1, \ldots, X_k$.
- β_0 is the **intercept** of the regression.
- u_i is the random **error** term.

The Multiple Linear Regression. Matrix notation

Sometimes it is useful to express the MLR model using matrix notation:

$$\boldsymbol{Y} = \beta \boldsymbol{X} + \boldsymbol{u},$$

where

$$\boldsymbol{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad \boldsymbol{X} = \begin{pmatrix} 1 & X_{11} & \cdots & X_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \cdots & X_{nk} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \boldsymbol{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}.$$

- $\boldsymbol{Y}$ is the response vector
- $\boldsymbol{X}$ is the *design matrix*
- β the coefficient vector
- $\boldsymbol{u}$ is the random error vector

The OLS estimators in the MLR

The **Ordinary Least Squares (OLS) estimators** of $\beta_0, \ldots, \beta_k$, denoted by

$$\widehat{\beta}_0, \ldots, \widehat{\beta}_k,$$

are those values of the model's coefficients that minimizes the (distance) function

$$D(\beta_0, \ldots, \beta_k) := \sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 X_{1i} - \cdots - \beta_k X_{ki})^2 = \sum_{i=1}^{n} u_i^2.$$

It can be proved that the OLS vector estimator $\widehat{\boldsymbol{\beta}} = (\widehat{\beta}_0 \cdots \widehat{\beta}_k)^T$ takes the form

$$\widehat{\boldsymbol{\beta}} = (\boldsymbol{X}^{\boldsymbol{T}} \boldsymbol{X})^{-1} \boldsymbol{X}^{\boldsymbol{T}} \boldsymbol{Y}.$$

In the simple linear case:

$$\widetilde{\beta}_1 = \frac{\mathbb{C}ov\,[Y, X]}{\mathbb{V}ar\,[X]}, \quad \widehat{\beta}_0 = \bar{Y} - \widehat{\beta}_1 \bar{X}.$$

Two core assumptions of the MLR and unbiasness of the OLS estimators

Introduce the assumptions

L1 **Linearity:** $Y = \beta X + u$

L2 **Zero conditional mean:** $\mathbb{E}\left[u \mid X\right] = 0$
The value of the regressors must contain no information about the mean of the unobserved factors

Under assumptions $L1$ and $L2$, the OLS estimators $\widehat{\beta}$ is unbiased, meaning that

$$\mathbb{E}[\widetilde{\beta}] = \beta$$

Interpretation of the slope coefficients

Let $\Delta_j Y$ represents the increment in the expectation of Y given the regressors, by changing the jth regressor ΔX_j units and keeping constant the other regressors. Then, under Assumptions L1-L2, $\Delta_k Y = \beta_k \Delta X_k$, that is

$$\beta_k = \frac{\Delta_k Y}{\Delta X_k}.$$

Therefore,

Interpretation of the the slope coefficients

Therefore, β_j is the average effect that increasing X_j one unit has on the response Y, holding the other regressors fixed (under **ceteris-paribus**)

Fitted values and residuals

Once we have the least squares estimates, we can define the next two concepts:

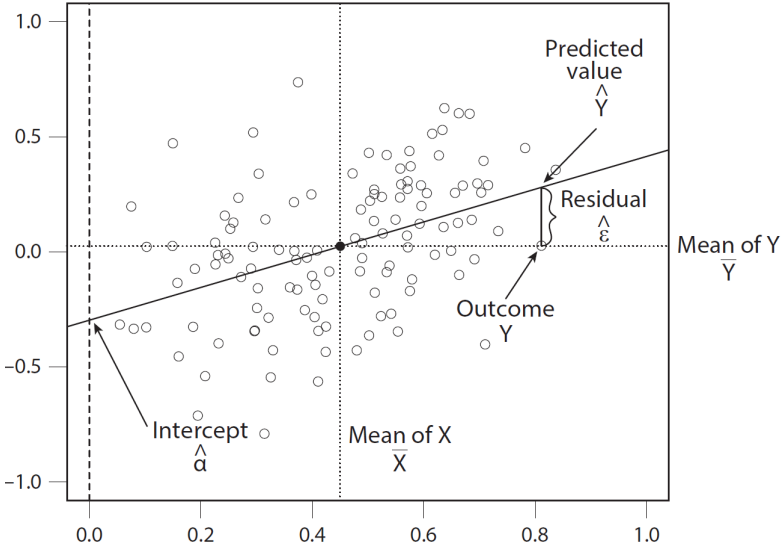
- The (OLS) **predicted (or fitted) values** are

$$\widehat{Y}_i := \mathbb{E}\left[Y_i|X\right] = \widehat{\beta}_0 + \widehat{\beta}_1 X_{1i} + \cdots + \widehat{\beta}_k X_{ki}.$$

- The **residuals** are

$$\widehat{u}_i = \widehat{Y}_i - Y_i.$$

Illustrative figure in the simple linear case



Interpretation of the OLS estimators, fitted values, and residuals

- **The OLS estimators are estimators** of the unknown population parameters $\beta_0, \ldots, \beta_k$
- **The fitted values are estimators** of the mean of the response given the predictor, that is, $\mathbb{E}[Y|X]$
- **The residuals are estimators** of the error term

Measures of fit

Measures of fit in the MLR

Three commonly used summary statistics in multiple regression are

- the **Standard Error of the Regression** (SER)
- the **coefficient of determination of the regression** (R^2)
- the **adjusted R^2** ($\bar{R}^2$)

All three statistics measure how well the OLS estimate of the multiple regression line describes, or "fits", the data.

The Standard Error of the Regression

The standard error of the regression (SER) **estimates the standard deviation of the error term** u_i

The SER **is a measure of the spread of the distribution of** Y **around the regression line**

In multiple regression, the SER is defined as

$$\text{SER} = \sqrt{\frac{\text{SSR}}{n-k-1}},$$

where k is the number of predictors in the regression and SSR is the Sum of Squared Residuals, that is,

$$\text{SSR} = \sum_{i=1}^{n} \widehat{u}_i^2.$$

The R^2

The regression **R^2 is the fraction of the sample variance of Y_i explained by (or predicted by) the regressors:**

$$R^2 = \frac{\text{ESS}}{\text{TSS}},$$

where ESS is the Explained Sum of Squares and TSS is the Total Sum of Squares, that is,

$$\text{ESS} = \sum_{i=1}^{n} (\widehat{Y}_i - \overline{Y})^2, \quad \text{TSS} = \sum_{i=1}^{n} (Y_i - \overline{Y})^2.$$

Equivalently, since $\text{TSS} = \text{SSR} + \text{ESS}$, the R^2 is 1 minus the fraction of the variance of Y_i not explained by the regressors:

$$R^2 = 1 - \frac{\text{SSR}}{\text{TSS}}.$$

The misguidance of the R^2. The adjusted R^2

The R^2 increases whenever a regressor is added, unless the estimated coefficient on the added regressor is exactly 0, which is practically impossible (probability 0).

The increase in the R^2 after adding a new variable does not mean an actual improvement in the fit of the model. .

In this sense, **the R^2 gives an inflated estimate of how well the regression fits the data in the MLR model**.

To correct this effect, the corrected R^2, or **$\overline{R}^2$, is defined by deflating or reducing the R^2 by some factor:**

$$\overline{R}^2 = 1 - \frac{n-1}{n-k-1}\frac{SSR}{TSS} = 1 - \frac{n-1}{n-k-1}(1 - R^2),$$

where n is the sample size and k is the number of regressors.

The $\overline{R}^2$ does not necessarily increase when a new regressor is added.

Prediction

In-Sample Predictions in Linear Models

In-Sample Prediction:

- Predictions made for the same data points used to fit the model.
- Also called fitted values.
- Assesses how well the model fits the training data.
- Evaluated using metrics like R^2, Mean Squared Error (MSE), or residuals.
- **Example:** Predicting house prices using the dataset used to train the model.

Out-of-Sample Predictions in Linear Models

Out-of-Sample Prediction:

- Predictions made for new, unseen data points not used during training.
- Measures the model's generalizability to other datasets.
- Typically evaluated using validation or test datasets.
- **Example:** Using a house price model trained on one dataset to predict prices in a different city.

Within-Range Predictions in Linear Models

Within-Range Prediction:

- Predictions made for data points within the range of predictor values observed during training.
- Outside range the model's assumptions may not hold.
- Provides reliable estimates since the model is not extrapolating.
- **Example:** Predicting house prices for properties with characteristics similar to (withing the range of) the training data.

An example: facial appearance and election outcomes

Does facial appearance predict election outcomes?[1]

Case Study!

Which person is the most competent?

[1] Alexander Todorov, Anesu N. Mandisodza, Amir Goren, and Crystal C. Hall (2005) Inferences of competence from faces predict election outcomes.

Nonlinear models

Nonlinear models

The linear model is termed linear not because the regression curve is a line, but because **the effects of the parameters β_0 and β_1 are linear.**

Indeed, **the predictor X may exhibit a nonlinear effect on the response Y and still be a linear model!**

For example, the following models **can be transformed into simple linear models**, by considering the relation between Y and the new regressor $\widetilde{X}$:

- $Y = \beta_0 + \beta_1 X^2 + u, \quad \widetilde{X} = X^2$
- $Y = \beta_0 + \beta_1 \ln(X) + u, \quad \widetilde{X} = \ln(X)$
- $Y = \beta_0 + \beta_1 (X^3 + \ln(|X|) + X^2) + u, \quad \widetilde{X} = X^3 + \ln(|X|) + X^2$

Nonlinear models

This "linearization trick" is a simple way to **extend enormously the functionality of the linear model.**

Indeed, the new transformed regression takes the form

$$Y = \beta_0 + \beta_1 f(X) + u,$$

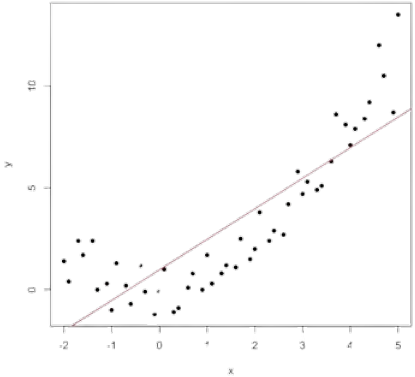
where f could be any function whose domain admits the values of the regressor.

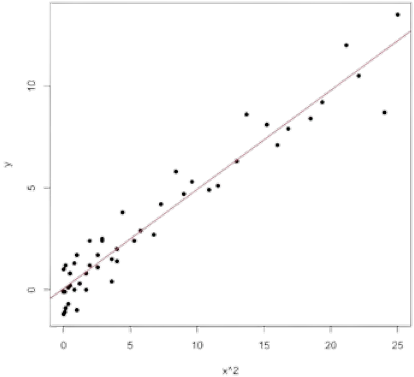
This is **particularly useful in real applications**, where linearity is hardly verified.

Some of **the most commonly used transformations** are:

- (Polinomial) $f(x) = x^k, \ k \in \mathbb{N}$
- (Roots) $f(x) = x^{1/k}, \ k \in \mathbb{N}$
- (Exponential) $f(x) = e^x, \ f(x) = e^{-x}$
- (Logarithmic) $f(x) = \ln(x)$

Quadratic pattern Y vs X (left) and Linearized pattern Y vs X^2 (right)





You can visualize more transformations here

Nonlinear models. General case

We can also transform the response to achieve linearity,

In such a case, **the general transformed linear model** takes the form

$$g(Y) = \beta_0 + \beta_1 f(X) + u,$$

for two suitable functions f and g.

It is important for g to be one-to-one, to be able to explain the original variable Y and not the transformation $g(Y)$.

The linear model for causal inference under randomized control trials

RCTs and Linear Regression Causal Inference: the binary (dummy) regressors case

- For the sake of simplicity, let us consider a simple setting to illustrate the concept:

$$Y = \beta_0 + \beta_1 X + u.$$

- Y is the response
- X is the binary treatment ($X = 1$ for treatment and $X = 0$ for control)
- When X is binary,

$$\widehat{\beta_0} = \underbrace{\frac{1}{n_0}\sum_{i=1}^{n}(1 - X_i)Y_i}_{\text{mean outcome for control}}$$

$$\widehat{\beta_1} = \underbrace{\frac{1}{n_1}\sum_{i=1}^{n}X_i Y_i}_{\text{mean outcome for treated}} - \underbrace{\frac{1}{n_0}\sum_{i=1}^{n}(1 - X_i)Y_i}_{\text{mean outcome for control}} .$$

- Therefore, $\widehat{\beta_1}$ is the **difference-in-mean estimator** of the **estimated average treatment effect (ATE)**.

Linear regression as a difference-in-means estimator

Linear regressions equivalence to difference-in-means estimators

When applied to **experimental data** with a single, **binary treatment**, **the estimated slope coefficient of the linear regression model can be interpreted as an estimate of average treatment effect and is numerically equivalent to the difference-in-means estimator**.

The estimated intercept, on the other hand, is equal to the estimated average outcome under the control condition. The randomization of treatment assignment permits this causal interpretation of association identified under a linear regression model.

Case Study. Simple regression

Do female politicians promote different policies than male politicians?[1]

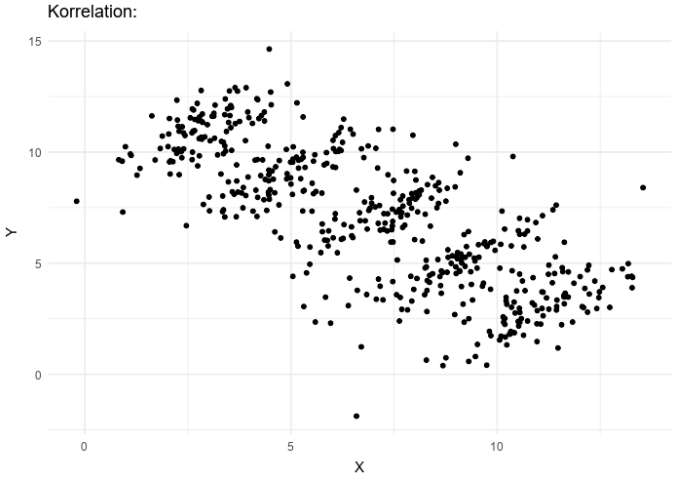
Case Study!

[1] Raghabendra Chattopadhyay and Esther Duflo (2004) Women as policy makers: Evidence from a randomized policy experiment in India.

Heterogeneous treatment effect

- **Heterogeneous treatment effect** refers to the variation in the impact of a treatment across different individuals or groups (e.g., effect of drug between patients in different age ranges).
- **The Simpson paradox:** when a trend observed in multiple groups changes when the groups are aggregated.
- **Linear regression with multiple predictors** can help uncover these variations by identifying characteristics associated with differences in the treatment effect's direction and magnitude.
- **Understanding** heterogeneous treatment effects is crucial for tailoring treatments to those who will benefit most and avoiding adverse outcomes for others.

The Simpson Paradox



Simpson Paradox!

Introduction to Linear Regression with Interaction Terms

Goal: Model how the effect of a predictor (treatment) varies based on another (covariate) variable.

Example of linear regression with interaction:

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 (X \cdot Z) + u$$

- X: Treatment variable
- Z: Covariate variable (e.g., age, gender, income)
- $X \cdot Z$: **Interaction term** capturing the interaction effect
- β_3: Coefficient of interaction term, indicating how the effect of X changes with Z

Interaction coefficient interpretation. An example

Example of linear regression with interaction:

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 (X \cdot Z) + u \tag{1}$$

Interpretation of the interaction term

The linear model (1) with the interaction term $\beta_3(X \cdot Z)$ assumes that the effect of X linearly depends on Z. That is, as we increase Z by one unit, the change in the average outcome associated with a one-unit increase of X goes up by β_3.

Interaction terms may also be non-linear, like $(X \cdot Z^2)$ or $(X \cdot \ln(Z))$.

Case Study. Heterogeneous treatment effect. Categorical variables

Do social pressure increases participation in elections?[1]

Case Study!

[1] Alan S. Gerber, Donald P. Green, and Christopher W. Larimer (2008) Social pressure and voter turnout: Evidence from a large-scale field experiment.

The linear model for causal inference under observational studies

Regression discontinuity design. Definition.

Regression Discontinuity Design (RDD)

Quasi-experimental method used to estimate causal effects when a treatment is assigned based on a cutoff or threshold in a continuous variable.

- **Key Assumption:** Units just above and below the cutoff are similar in all respects except for the treatment received.
- **Assignation mechanism:** Observational units are assigned a score (e.g., income, age, or test scores), and treatment is assigned when this score crosses a predetermined threshold.
- **Applications:** Common in policy evaluation, such as the impact of scholarships awarded based on test scores or voting outcomes based on a minimum percentage.
- **Validity** Often has strong internal validity, but it may lack external validity because the result may not be generalizable to observations away from the point of discontinuity.

Regression discontinuity design. Two regressions

Regressions:

$$Y_i = \beta_0^{\text{control}} + \beta_1^{\text{control}} X_i + u_i, \qquad \text{for } i \text{ such that } X_i < c,$$
$$Y_i = \beta_0^{\text{treat}} + \beta_1^{\text{treat}} X_i + u_i, \qquad \text{for } i \text{ such that } X_i \geq c.$$

- Y: Outcome variable.
- X: Regressor, determines treatment assignment.
- c: Cutoff threshold.

ATE estimation:

$$\begin{aligned}\lim_{X \to c^+} E[Y|X] - \lim_{X \to c^-} E[Y|X] &= (\beta_0^{\text{treat}} + \beta_1^{\text{treat}} c) - (\beta_0^{\text{control}} + \beta_1^{\text{control}} c) \\ &= (\beta_0^{\text{treat}} - \beta_0^{\text{control}}) + c(\beta_1^{\text{treat}} - \beta_1^{\text{control}})\end{aligned}$$

Regression discontinuity design. One regression

Regression: $Y = \beta_0 + \beta_1 X + \beta_2 T + \beta_3 T \cdot X + u$

- Y: Outcome variable.
- X: Regressor, determines treatment assignment.
- T: Treatment indicator ($T = 1$ if $X \geq c, 0$ otherwise).
- c: Cutoff threshold.

ATE estimation:

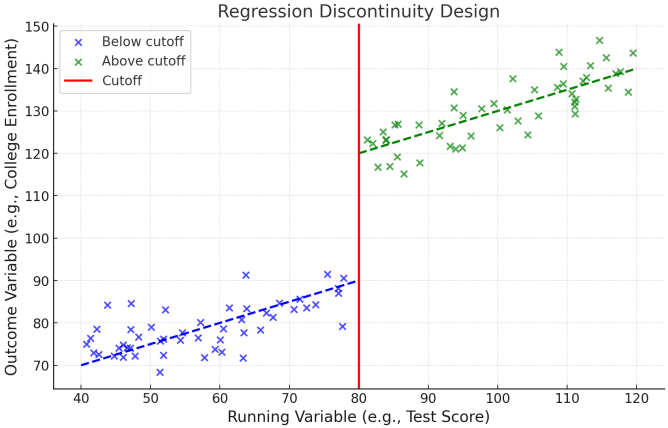
$$\lim_{X \to c^+} E[Y|X] - \lim_{X \to c^-} E[Y|X] = (\beta_0 + \beta_1 c) - (\beta_0 + \beta_1 c) = \beta_2 + \beta_3 c$$

Connection with the two-regressions case

$$\beta_0 = \beta_0^{\text{control}}, \quad \beta_1 = \beta_1^{\text{control}}, \quad \beta_0 + \beta_2 = \beta_0^{\text{treat}}, \quad \beta_1 + \beta_3 = \beta_1^{\text{treat}}$$

Regression discontinuity design. Visualization

Example: scholarships are awarded to students scoring above 80 on a test. Plot average test scores on the x-axis and the likelihood of college enrollment on the y-axis, showing a clear jump at the cutoff.



Can politicians increase their personal wealth due to holding office?[1]

Case Study!

[1] Andrew C. Eggers and Jens Hainmueller (2009) MPs for sale? Returns to office in postwar British politics.